jwselect: Quick User Guide

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Select optional preperiod restrictions and estimate treatment effects by minimum distance (MD), using cohort–time estimates and their joint covariance. This guide covers ordinary estimation, including full nonlinear conditional-index blocks in version 0.7.0, alongside Sun–Abraham, CS-DR, and the existing adapters.

1 Install

The main command requires Stata 16.1 or later. Extract the current archive and replace the path below with the folder containing `jwselect.pkg`:

```
net install jwselect, from("C:/path/to/jwselect") replace
which jwselect
findfile _jwselect_mata.do
help jwselect
```

After an update, save your work and restart Stata. To use `jwselect` directly after `jwdid`, first install its dependencies:

```
ssc install require, replace
ssc install ftools, replace
ssc install reghdfe, replace
ssc install hdfe, replace
net install jwdid, ///
    from("https://raw.githubusercontent.com/friosavila/stpackages/main") replace
```

The matrix interface requires no upstream estimator. The CS-DR adapter requires `csdid` 2.x, version 2.0.1 or later. The README gives the pinned 2.0.1 install command; consult the authors' installation guide when updating an existing installation. `jwselect` does not install dependencies. Run commands containing `///` together from the Do-file Editor. An optional installation check, in a fresh Stata 17 session, is `do run_release_tests.do` from the extracted package folder. The test clears data and estimates; save your work first.

2 A typical analysis

Replace the variable names below with yours. Code never-treated units as `first_treat=0`; treated units carry their first treatment date.

```
jwdid y x1 x2, ivar(id) tvar(year) gvar(first_treat) ///
    never method(regress) cluster(id)
estimates store JW_SOURCE

* Maintained-baseline estimate, with no optional restrictions
jwselect, select(manual) postwindow(0 2)
estimates store JW_BASE

* Default: cohort-wise Wald-BIC, followed by joint MD
estimates restore JW_SOURCE
jwselect, select(wbic) candidates(nested) postwindow(0 2)
estimates store JW_WBIC
```

Restore `JW_SOURCE` before every new direct `jwselect` run. Each successful run replaces Stata's estimation results. A bare `jwselect` only redisplays the current results. Here, `postwindow(0 2)` averages

the first three treatment periods within each cohort.

Supported linear input. Use a balanced, regularly spaced panel, unweighted linear estimation, full cohort–time effects, clustering by `id`, and numeric covariates constant within each unit. The default `vce(unconditional)` includes covariate-averaging uncertainty and requires `method(regress)`. For other designs, use the matrix interface on page 7.

3 Sun–Abraham

Install optional `eventstudyinteract`, `avar`, `reghdfe`, and `ftools`; the README gives commands. Use the wrapper so the full cohort/event map, omitted baseline, and original sample can be verified:

```
jwselect_sa y, ivar(id) tvar(year) gvar(first_treat)
estimates store SA_SOURCE
jwselect, select(wbic) wbicscope(joint) postwindow(0 2)
estat selection
estimates restore SA_SOURCE
jwselect, select(manual) postwindow(0 2)
```

Code never-treated units zero. The supported design is a balanced, unweighted panel with no covariates and never-treated controls, clustering by unit, and positive integer unit-spaced time. The wrapper supplies all observed event indicators, omits only event -1, and retains the full regression covariance. Use `vce(supplied)`, the default for this source. Existing manual, Wald–BIC, and focused selection are available; linear baseline rebasing is supported. Raw unverified `eventstudyinteract` fits are not accepted.

Fixed aggregation. These results use fixed equal-cohort, equal-cell, or custom weights. They do not implement population IW uncertainty from estimated cohort shares. See `help jwselect_sa` and the full guide for this distinction and the required data checks.

Monte Carlo check. The new do-file compares baseline, all-preperiod, joint Wald–BIC, and aggregate-focused estimates under valid, local-violation, and fixed-violation designs. It records bias, RMSE, and coverage, including focused reselection intervals and an uncorrected normal benchmark. From the extracted package root:

```
do examples/jwselect_sa_monte_carlo.do smoke 2 600
do examples/jwselect_sa_monte_carlo.do full 1000 1000
```

Save work first. A new results folder contains replication data, summaries, settings, and optimization diagnostics. The first run checks execution; the second is a computationally demanding coverage experiment. Inspect failure counts and numerical flags before interpreting coverage. The full guide gives the design and optional directory arguments.

4 CS-DR and nonlinear DiD

CS-DR: start from unaggregated `csdid` results

```
csdid y x1 x2, ivar(id) time(year) gvar(first_treat) ///
    method(dr) nevertreated base_period(universal) ///
    analytical pointwise cluster(id)
estimates store CS_SOURCE
jwselect, select(wbic) postwindow(0 2)
estat selection
estimates restore CS_SOURCE
jwselect, select(manual) postwindow(0 2)
```

Use an unweighted balanced panel with positive integer, unit-spaced time, never-treated units, and plain numeric covariates constant within unit. Clustering can be omitted or specified at `ivar()`; higher-level

clusters are unsupported. Retain the original estimation data. Older csdid 1.x, aggregated results, bootstrap covariance, varying bases, and not-yet-treated controls are rejected. vce(supplied) is the default and only covariance convention for this source; it retains the full analytical joint covariance.

Nonlinear DiD: full conditional-index blocks

```
jwdid y x1 x2, ivar(id) tvar(year) gvar(first_treat) ///
    never method(logit) cluster(id)
estimates store NL_SOURCE
jwselect, restrictions(index) select(wbic) postwindow(0 2)
estat design
estat pretrend
estimates restore NL_SOURCE
jwselect, restrictions(index) select(manual) postwindow(0 2)
```

Use method(poisson) for an exponential conditional mean. Both methods require the balanced, unweighted, never-treated, unit-clustered, time-invariant-covariate design. restrictions(index) selects each lead's intercept and all covariate interactions together. With two covariates, selecting one date imposes three zero restrictions; Wald–BIC counts all three. Candidate families enumerate dates, and prematrix() selects complete date blocks. Default centered interactions and upstream xasis are supported. Unidentified or singular blocks are rejected.

The target remains an outcome-scale ATT. A common unit-score calculation supplies its full covariance with the index coefficients. Default vce(unconditional) includes covariate-averaging uncertainty; vce(conditional) conditions on the observed covariate distribution. estat design reports the block ranks and coefficient map. Omitting restrictions() preserves the earlier restrictions(placebo) average-placebo route. A zero average placebo is weaker than a zero full index block when effects cancel across covariates.

Keep the source baseline and interpret the adjustment

CS-DR and nonlinear adapters require baseline(cohort). Omitted anticipation() is inferred from the source; an explicit mismatch is rejected. For jwdid, it is the recorded anticipation minus one. Restore the source before every new specification.

For a selected set A, the target-level MD estimate is $\widehat{d} - \widehat{V}_{dh_A}\widehat{V}_{h_Ah_A}^{-1}\widehat{h}_A$. Here h_A contains full index blocks under restrictions(index), or average placebo moments for CS-DR and restrictions(placebo). The index route selects complete lead blocks, uses restriction rank in WBIC, and computes target-level MD. It does not perform a constrained nonlinear refit or the separate coefficient-MD prediction estimator. Adjusted ATTs are not clipped to natural bounds. The relevant identifying and covariance conditions must hold, and selection needs the inference qualifications below. Fixed aggregation weights need not match csdid's defaults. See examples/nonlinear_index_demo.do, examples/nonlinear_demo.do, and examples/csdid_demo.do.

5 Choose the design and preperiod restrictions

Baseline and treatment-effect horizon

The baseline is maintained, never selected or pretrend-tested. Optional dates add restrictions. Common-baseline choices below apply to linear, Sun–Abraham, and matrix inputs; CS-DR and nonlinear adapters retain their upstream cohort baseline. For annual data, g is the cohort's treatment date and q the earliest input treatment date. With baseline(common) and no anticipation, eligible dates extend through $g-1$, excluding the baseline $q-1$.

Option	Meaning
`baseline(cohort)`	Baseline $g-1$; the default.
`baseline(common)`	Baseline $q-1$ for every cohort.
`anticipation(1)`	Allow one anticipation period: use $g-2$ or $q-2$ and exclude potentially affected optional dates. Default: 0 for linear, SA, and matrix inputs; inferred for CS-DR and nonlinear inputs.
`postwindow(0 0)`	Immediate effect only, for each cohort.
`postwindow(0 2)`	Average over event times 0, 1, and 2.

Without `postwindow()`, all available post-treatment cells are used. Every requested date must exist for every cohort. For spacing $h =$ `gap()` and anticipation δ, the baseline is $g-h(1+\delta)$ or $q-h(1+\delta)$; event time is $(t-g)/h$. Direct input obtains spacing from the source; the CS-DR adapter currently requires unit spacing. The two commands use different anticipation conventions; do not copy the `jwdid` anticipation value mechanically into `jwselect`.

Baseline only, manual choice, or automatic selection

Option	What it does
`select(manual)`	Use only the maintained baseline unless `prematrix()` is supplied.
`select(wbic)`	Default. Select within each cohort; add `wbicscope(joint)` for joint selection (below).
`select(focused)`	Balance estimated bias and variance for a chosen target; see page 5.

For manual selection, list optional dates in a two-column matrix: *cohort, calendar time*. For example, if cohorts 4 and 6 and these dates are present in your data:

```
matrix P = (4,1 \ 4,2 \ 6,3 \ 6,4)
estimates restore JW_SOURCE
jwselect, select(manual) prematrix(P) postwindow(0 2)
```

Do not list the baseline. Dates need not be consecutive. Manual and Wald–BIC use the *union* of the chosen restrictions in joint MD: one cohort's restrictions can affect other estimates through covariance. A cohort absent from `P` contributes no optional restrictions of its own.

Choose the candidate family explicitly

With optional dates $\{1,2,3\}$, `candidates(nested)` searches $\varnothing, \{3\}, \{2,3\}, \{1,2,3\}$. `candidates(all)` searches all eight subsets, including $\{1,3\}$. With K optional dates, these families have $K+1$ and 2^K candidates. The package default is `nested`; request `all` explicitly for exhaustive search. Baseline only is included in both families.

The limits are `maxcand(4096)` per cohort and `maxjoint(4096)` for a joint WBIC or aggregate-focused family. Exceeding a limit causes an error; no candidates are silently dropped. All-subset searches grow quickly; raise a limit only if you intend the larger family.

Aggregation. The default gives cohorts equal weight, divided equally among their included post periods. `aggregate(equalcell)` weights included cells equally.

Joint Wald–BIC selection (new in 0.4.0)

Use one joint criterion to choose restrictions across all cohorts:

```
estimates restore JW_SOURCE
```

```
jwselect, select(wbic) wbicscope(joint) ///
    candidates(nested) postwindow(0 2)
estat selection
estat scores
estimates store JW_JOINT
```

The joint criterion is $W_A - r_A \log N$: the joint pretrend Wald statistic, minus the number of restrictions times the log of the independent-unit count. It uses covariance across cohorts and dates, then estimates all effects by joint MD. The default `wbicscope(cohort)` selects within cohorts before joint MD; these two rules can choose different sets. Optional date-priority penalties are not implemented; `aggweights()` changes the target, not the WBIC reward.

The joint family combines every local candidate with every candidate in other cohorts. With K_g optional dates in cohort g, there are $\prod_g (K_g + 1)$ nested configurations or $2^{\sum_g K_g}$ all-subset configurations. Even nested joint search may exceed `maxjoint(4096)`. A prespecified `jointmatrix(JM)` can supply a smaller family: one binary joint mask per row, with columns in `e(hmap)` order. Obtain that map with `estat design`; see `help jwselect` for details. Fix the family before inspecting outcomes. Raise a cap only for a computationally feasible family.

Joint pretrend tests

After any successful `jwselect` fit:

```
* All eligible optional preperiod restrictions, jointly
estat pretrend
return list

* Manual/Wald-BIC selected set: diagnostic after selection
estat pretrend, selected
```

For focused results, explicitly choose an output instead:

```
matrix list e(output_map)
estat pretrend, selected output(1)
```

`output()` means a result-row number and is allowed only with `selected`. Manual/Wald–BIC selected tests default to row 1, since all rows share the same joint restriction set. For a prespecified subset, use `estat pretrend, prematrix(P)`; `P` has the same two-column cohort/date format as manual selection. Do not combine it with `selected`.

The test reports Wald, the equivalent MD J, degrees of freedom, and a chi-square reference p-value. J is evaluated through the Wald identity under the compatible covariance-weighted MD model, not by an independent refit. `r(table)` has columns `df Wald J pvalue`; separate scalars include `r(chi2)`, `r(J)`, `r(df)`, and `r(p)`.

Interpretation. A prespecified test uses the chi-square reference under the maintained model. A p-value for an outcome-selected set is a diagnostic, not an automatic correction for selection. Rejection concerns optional restrictions; the maintained baseline is never tested. For a focused aggregate of cohort-selected effects, the test uses the union of their restrictions; J refers to that set's compatible joint MD model. An empty set has zero degrees of freedom and no p-value.

6 Focused estimates and confidence intervals

Request cohort-focused estimates, their aggregate, and numerical inference:

```
estimates restore JW_SOURCE
jwselect, select(focused) focus(cohort) ///
    candidates(all) postwindow(0 2) level(95)
estat inference
```

Option	Selection target and reported estimates
`focus(cohort)`	Select for each cohort's mean effect; also report the aggregate of the resulting cell estimates.
`focus(aggregate)`	Select a joint restriction configuration for the aggregate. Cohort results use that same selected model.
`focus(both)`	Package default. Report cohort-focused effects, their aggregate, and a separately aggregate-focused estimate.

For point estimates only, add `inference(none)`; intervals and p-values will be missing.

How to read focused inference

The default `inference(auto)` repeats selection inside joint Gaussian draws and searches over plausible preperiod violations. Use the reported intervals and p-values. `se_fixed` is a fixed-selection diagnostic; do not build ordinary normal intervals from it. Intervals are marginal for each effect, not simultaneous bands, and need not be narrower than baseline-only intervals. The focused-effect consistency result does not establish validity of the selected preperiods, uniform MSE improvement, or interval coverage. Report the baseline estimate and interval alongside selected results.

Search budget: the controls that matter

Option	Default	Meaning
`optmaxeval()`	400	Evaluations per objective, shared across starts.
`optstarts()`	4	Starting points per objective.
`optreps()`	1,999	Common Gaussian draws used during search.
`reps()`	19,999	Independent draws used to check discovered points.
`opttol()`	0.02	Stopping mesh size in standardized coordinates.
`seed()`	12,345	Reproducible simulation seed.

There are four objectives per effect: two interval bounds and two p-value tails. Each start receives 100 evaluations by default. **"Search budget exhausted" means at least one start reached its allowance.** The command returns the best points found; this is not a Stata failure. Inspect `budget_exhausted` and `converged_starts` in `estat inference`.

First increase the evaluation budget while retaining the same design:

```
estimates restore JW_SOURCE
jwselect, select(focused) focus(cohort) ///
    candidates(all) postwindow(0 2) level(95) optmaxeval(1600)
estat inference
```

Compare intervals and p-values with the earlier run. More starts, more draws, and another seed provide further checks. Increasing `reps()` alone does not resolve budget exhaustion. Converged searches do not certify global extrema.

The defaults `gamma(.01)` and `mcerr(.001)` allocate error to the nuisance region and simulation checks. See the full guide before changing them.

7 Read, save, and supply results

Inspect and save a completed run

Each cohort row averages its included post periods. The aggregate uses the chosen cell weights. `e(output_map)` identifies the actual cohort behind each row; an aggregate has cohort value zero.

```
jwselect                    // Redisplay the current results
estat selection             // Candidate sets and selected restrictions
estat design                // Baselines and cohort-time maps
estat cells                 // Individual cohort-time estimates
matrix list e(output_map)
estimates save "jwselect_results.ster", replace
```

These commands follow a successful `jwselect` call. Store or save each result you want to compare before restoring `JW_SOURCE`.

Stored result	Use
`e(table)`	Main table: estimate, `se_fixed`, interval bounds, and zero-effect p-value.
`e(b)`, `e(ci)`, `e(pvalue)`	Estimates, confidence intervals, and p-values separately.
`e(selection)`	Restrictions used by each output; absent when no optional preperiods exist.
`e(V)`	Joint covariance for manual and Wald–BIC results.
`e(V_fixed)`	Fixed-selection covariance; diagnostic for focused estimates.

Source metadata record the estimator, version, method, model, and restriction scale. Inspect them with `ereturn list`. The source version is empty when unavailable, including `jwdid`; retain its `which` output separately.

Prespecify manual restrictions, candidate families, and target weights. Manual intervals treat the chosen restrictions as fixed. Wald–BIC intervals use a pointwise oracle approximation under the method's assumptions; they are not uniformly valid near violations. Focused results do not post an ordinary `e(V)`. Do not use `e(V_fixed)` with conventional `lincom` or `test` as if it adjusted for focused selection.

Specification sensitivity. Changing controls requires refitting the source estimates and joint covariance, then reselecting on a common target and candidate family. Recompute focused inference to compare its intervals. The accompanying paper's control-removal results establish point estimate and selected-date sensitivity, not stability of focused intervals. Its weighted, time-varying-control application requires validated matrix inputs; it is outside the direct adapters' contracts.

Alternative input: coefficient and covariance matrices

Use this route for separately validated cohort–time estimates and their full covariance, including designs outside the direct adapters. It requires neither `jwdid` nor `csdid`.

```
jwselect, bmat(B) vmat(V) cells(C) nunits(500) ///
    gap(1) select(wbic) candidates(all) postwindow(0 2)
```

Input	Required contents
`B`	Row or column vector of unpooled treated-cohort–time estimates.
`V`	Full sampling covariance of `B`, including covariance across cohorts and dates. Supply V, not NV.
`C`	Matching rows: *cohort, calendar time*. Include the normalization cells and only positive treated-cohort codes.
`nunits()`	Number of independent sampled units, used in the Wald–BIC penalty; replace 500 with your count.
`gap()`	Period spacing, with default 1.

Repeat the matrix arguments for each specification. The convention is `vce(supplied)`. Supply unpooled cohort–time estimates and a validated full covariance; retain structural-zero normalizations, but never substitute zeros for unidentified effects.